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Studia Podyplomowe
Głębokie Sieci Neuronowe - Zastosowania w Mediach Cyfrowych

Graph neural networks in a casual knowledge graphs.

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YYYYY

WARSAW 2026

Graph neural networks in a casual knowledge graphs.

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Słowa kluczowe: NFT, ChatGPT, Buzzword300

Graph neural networks in a casual knowledge graphs.

Abstract.

Summary in English.....

Keywords: ChatGPT, Buzzword300

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Contents

1. Introduction

Krótki opis dziedziny problemu.

Motywacja, dlaczego warto zająć się rozwiązaniem postawionego problemu.

Opis wkładu własnego - co zrobiono w ramach przygotowywania pracy dyplomowej. Np. przygotowanie zbioru danych, opracowanie i implementacja architektury modelu itp.

YOLO **redmon2016you**

List of Symbols and Abbreviations

GNN – Graph Neural Network
DAG – Directed Acyclic Graph
HCR – Hierarchical Correlation Reconstruction
ADR – Adverse Drug Reaction
AKI – Acute Kidney Injury
DILI – Drug-Induced Liver Injury
GAT – Graph Attention Network
SAGE – Graph SAmple and aggreGatE
RGCN – Relational Graph Convolutional Network
AUC – Area Under the Curve
AUPRC – Area Under the Precision-Recall Curve

2. Thesis objective

The aim of this thesis is to develop a graph neural network models that could be applied in two tasks: a) to predict the presence of hidden links in a causal graph, b) to predict the values of endpoints for entities based on tabular data represented as a causal graph.

The two tasks are complementary ... [we will add here how the two tasks relate to each other in medicine + picture that shows it] The models will be trained and evaluated on synthetic datasets, as well as on real-world datasets. The performance of the models will be compared with baseline methods, and the results will be analyzed to understand the strengths and limitations of the proposed approach.

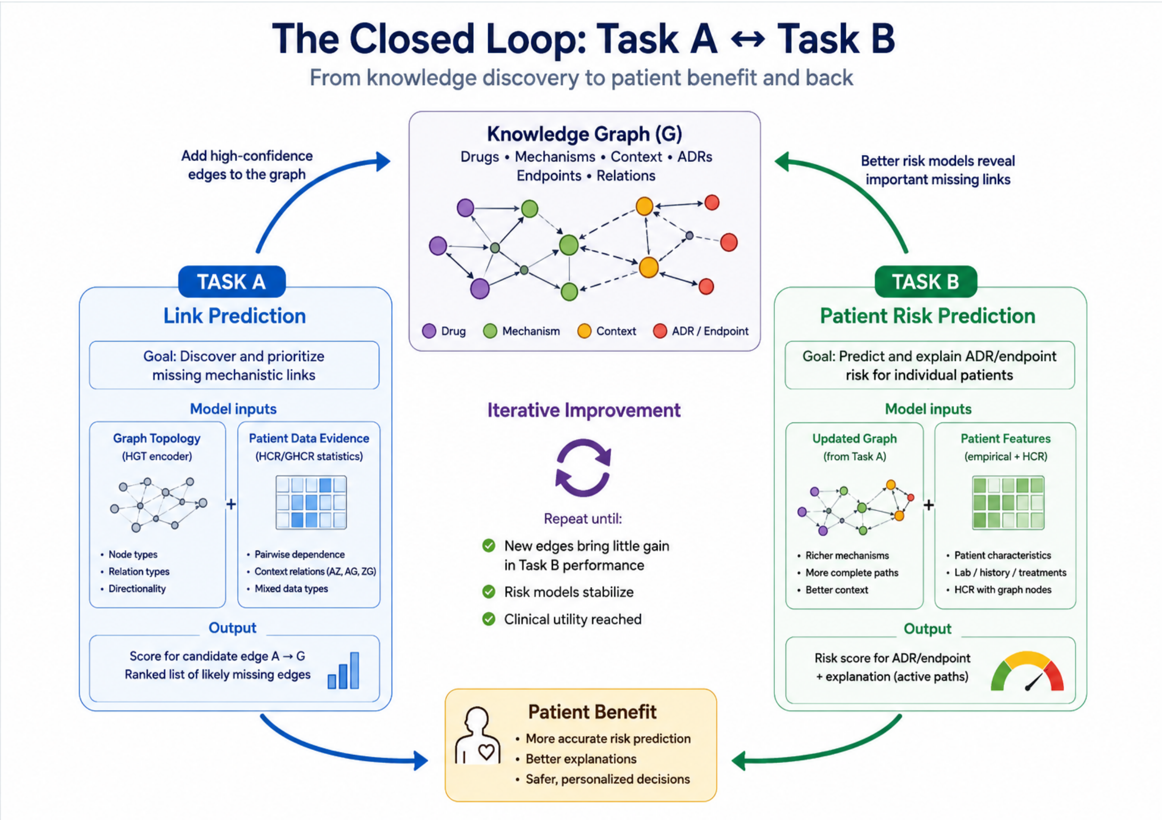

Figure 2.1. The description of the two tasks addressed in the thesis.

3. Related Work

Przegląd literatury związanej z rozwiązywanym problemem.

W pracy **he2016deep** opisano .... W **liu2022convnet** zaproponowano nowatorską architekturę.... Przykładowy rysunek 3.1.

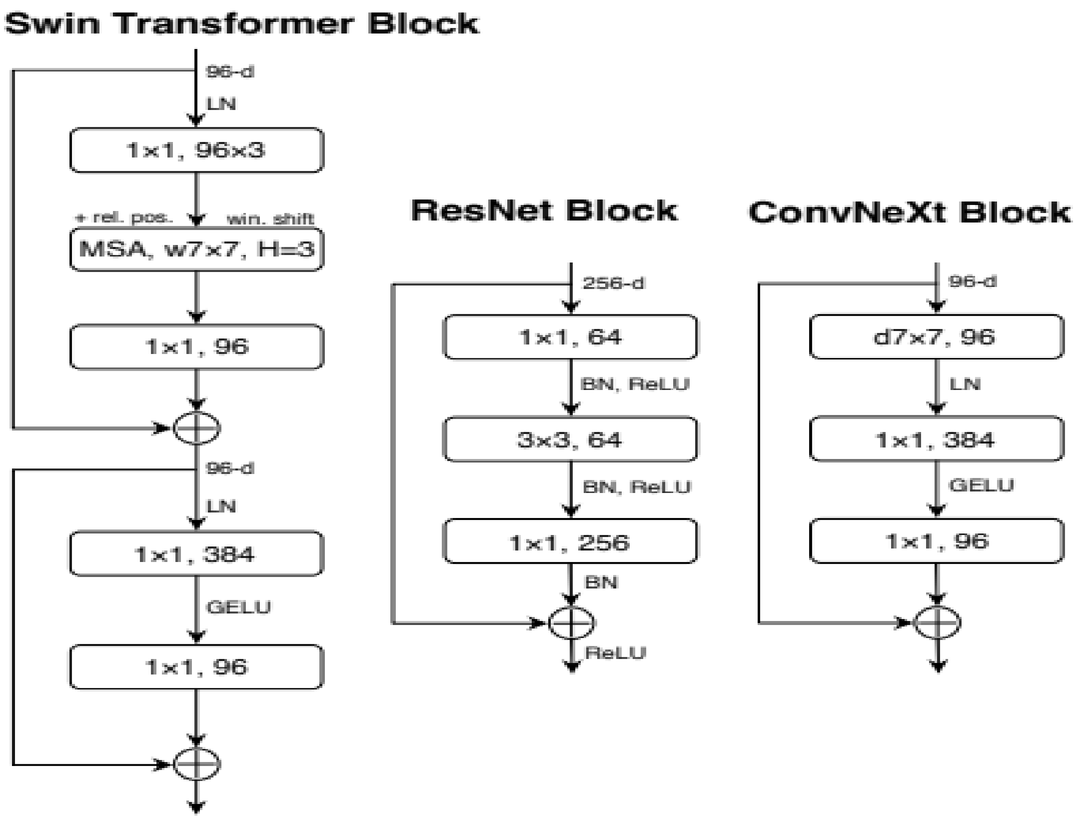

Figure 3.1. Porównanie bloków rezydualnych wykorzystywanych w sieci Swin Transformer, ResNet i ConvNeXt.

Position-aware (P-GNN)

Bayes sufit

3.1. Architecture

3.1.1. RGCN

3.1.2. Transformer

3.2. HCR

3.3. Hidden link reconstuction

3.4. Endpoint prediction

Endpoint prediction is a node classification tasks in data.

4. Dataset

4.1. Synthethic dataset

The dataset includes two structures - knowledge graph of medicine and patient dataset with pateint's characteristics.

4.1.1. Data generation

4.1.2. Knowledge graph description

Tabela 4.1. Node types in the patient causal DAG.

Node type	Description
`patient_context`	Stable patient-level covariates (e.g. demographics, comorbidities) available at the moment of prediction.
`drug_exposure`	Drugs the patient is exposed to; source nodes for pharmacotherapeutic causal paths.
`mechanism`	Intermediate physiological mechanisms mediating the effect of a drug or context on downstream states.
`adr_or_intermediate_state`	Intermediate adverse or physiological states positioned between mechanisms and clinical endpoints.
`observation_or_selection`	Nodes representing observation, documentation, or selection processes rather than a clinical state itself.
`clinical_endpoint`	Target adverse clinical outcomes (e.g. AKI, DILI, Hospitalization) predicted by the node classification head.

4.1.3. Patient data description

4.2. Input to node classification task

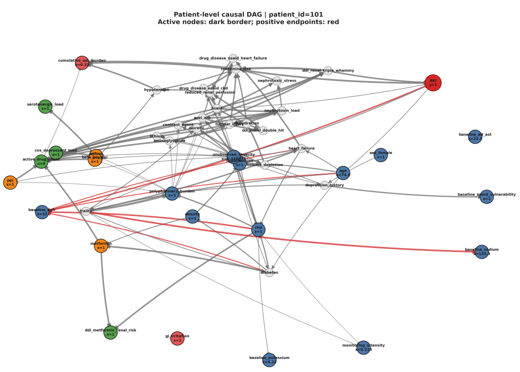

Figure 4.1. The description of the two tasks addressed in the thesis.

4.3. Proxy dataset

4.3.1. Last-FM dataset

4.3.2. Hetionet knowledge graph

5. Main Framework

A causal knowledge graph is not merely a static record of the literature: it can be systematically audited, weighted, and enriched with observational data, and subsequently used as the basis for explainable clinical prediction of drug-related outcomes. This thesis operationalizes that idea through two complementary graph neural network tasks, applied to a shared causal knowledge graph of pharmacotherapy:

a) **Task A:** link prediction, which reconstructs and updates the knowledge graph based on real patient data;

b) **Task B:** node classification, which predicts patient-specific clinical endpoints based on the knowledge graph and real patient data.

Since both tasks operate on the same underlying knowledge graph, we adopt graph neural networks as the common modeling framework, and enrich them with Hierarchical Correlation Reconstruction (HCR). In fig. 5.1 there is presented high-level approach to construct GNN with HCR. While the two tasks share this common foundation, their implementation details differ in several respects; we therefore present each task separately in the following subsections.

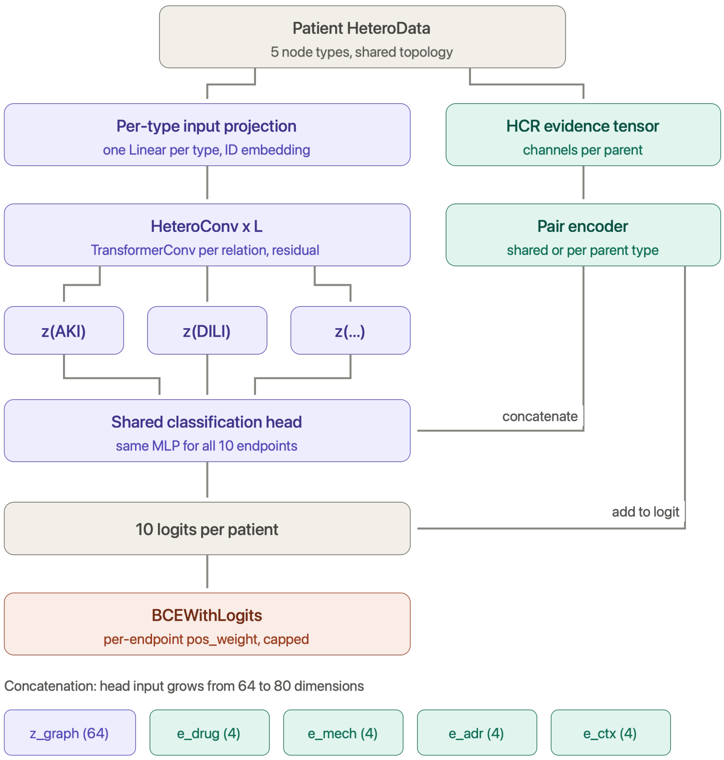

Figure 5.1. The presentation of approach to node classification task.

5.1. Link prediction in Graph Reconstruction

5.2. Endpoint prediction

Task B is formulated as multi-label node classification on a heterogeneous, per-patient graph. Rather than building one global graph and treating patients as additional nodes, every patient is represented as a copy of the same fixed causal DAG topology, with node feature vectors instantiated from that patient's individual covariates. The prediction targets are a fixed subset of clinical events on patient site. Formally, for a patient graph $G_p = (V, E)$ shared across the cohort, with patient-specific node features x_p, the model computes

$$\hat{y}_{p,e} = \sigma\big(f_\theta(G_p, x_p)_e\big), \quad e \in \mathscr{E}_{\text{target}}$$

where f_θ is a heterogeneous graph neural network shared across all patients, and only the leaf-level target embeddings are read out by the classification head.

5.2.1. Training Dataset - Data Representation

The first thing in applying GNNs is to built properly the input into neural network. The input graph for the node classification task was built based on:

a) **nodes**
b) **edges**
c) **per-scenario patient sample table**

Based on those items HeteroData object was created which was represented by objects type presented in the table

The patient data is encoded on nodes level. In fig. 5.1 there is presented the exemplary DAG for one of patient.

Two structural exclusion mechanisms are applied before any feature is computed:

a) nodes flagged `exclude_from_observed_model` or `is_latent` in the audit (hidden confounders);
b) nodes that are structural descendants of any endpoint in the DAG (e.g. documentation nodes such as `hospital_contact`, `fall_reported`, which are caused by the endpoint rather than causal parents of it)

are detected by graph descendant traversal and excluded from the feature set, even though they are not explicitly flagged in the source data. All normalization statistics (mean/std for continuous node values) and all HCR evidence statistics (section 5.2.3) are computed exclusively on the training split of patients, to prevent both node-level and patient-level information leakage across splits.

Tabela 5.1. Components of the per-patient `HeteroData` graph representation.

Component	Description
Node types	Six semantic categories of nodes shared across all patient graphs: `patient_context`, `drug_exposure`, `mechanism`, `adr_or_intermediate_state`, `observation_or_selection`, and `clinical_endpoint`.
Relations	Keyed by `(src_type, "to", dst_type)` weighted by `edge_attr` instead of using `edge_type`. This keeps the number of distinct convolutional relation modules in `HeteroConv` small while still letting edge-type information reach the message function.
Edge attributes (`edge_attr`)	A 1 + 3 + |edge types|-dimensional vector per relation: signed effect size, activation frequency, mechanistic confidence, evidence weight, and the edge-type one-hot. Convolutions consuming a scalar `edge_weight` (GraphConv/GCN) receive only the signed effect-size column; convolutions consuming a full `edge_attr` (GAT, Transformer) receive the whole vector.
Node features (`x`)	Static audited attributes plus a topological-layer encoding (`numeric` or `onehot`, configurable), combined with the patient's own observed value for that node where the node is visible under the current observability regime.

5.2.2. Model architecture

The core architecture is a stack of heterogeneous graph convolutional layers wrapped in HeteroConv, shared across relation types but parameterized identically for every patient graph. Four convolution backbones are supported through a common registry and were benchmarked against each other: GraphSAGE, GraphConv (GCN-style), GATv2, and TransformerConv. Because these operators were originally designed for homogeneous graphs, each is instantiated once per edge type inside HeteroConv, so the heterogeneous DAG is handled without collapsing node/edge types into a single homogeneous graph. As benefit of casual graph structure is possibility to encode deep structured path for node classification, multiple number of layers was tested to choose the best model.

A design issue that was handled during experiments is the duplication of the self-transform across relations. Several PyG convolution layers (SAGEConv, TransformerConv) internally add their own linear self-loop term, wrapped per-relation inside HeteroConv, this term is silently duplicated once per incoming relation type for a given node type. The implementation removes this internal duplication (`root_weight`=False for SAGE/Transformer) and replaces it with a single, explicit, controlled self-transform per layer per node type, applied once in the shared encoder rather than once per relation.

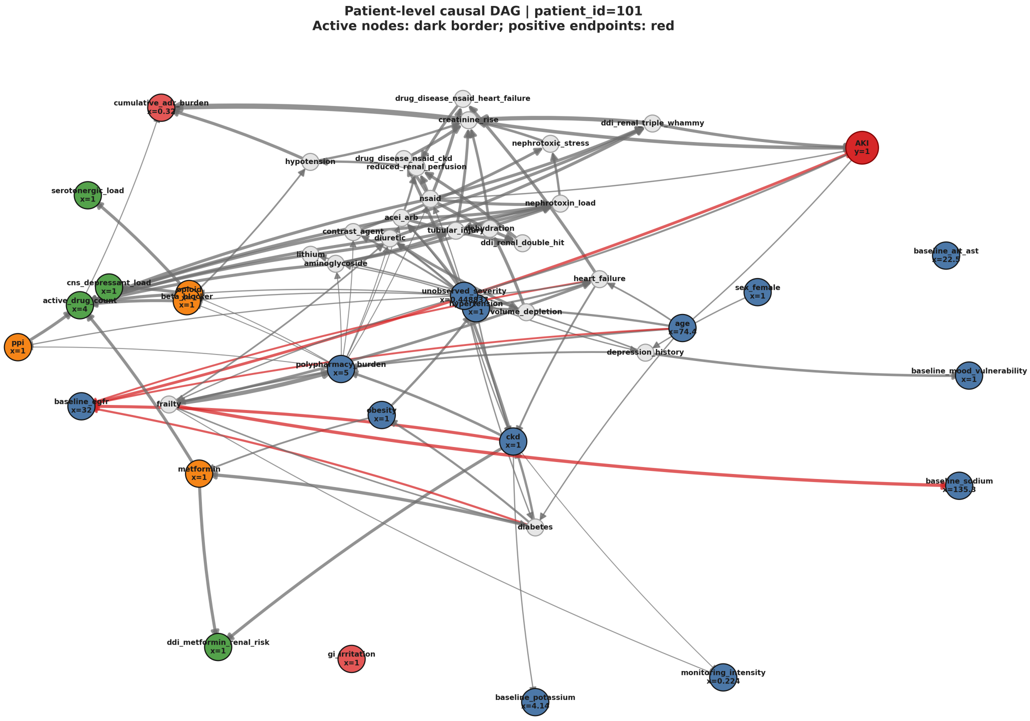

Figure 5.2. The description of the two tasks addressed in the thesis.

For GAT-family layers, where the self-dependency is baked into the attention query and cannot be disabled, the explicit self-transform acts as an additional controlled term on top of the irreducible built-in dependency. GraphConv/GCN, which has no `root_weight` parameter at all, is deliberately excluded from this mechanism, since its built-in self-transform cannot be removed without reimplementing the layer.

A node-identity embedding, added after the input projection, was introduced to resolve a specific representational collapse. Message-passing GNNs are bounded in expressive power by the 1-dimensional Weisfeiler-Lehman graph isomorphism test: two nodes with identical initial features and structurally equivalent neighborhoods cannot be assigned distinct representations by any number of convolutional layers. In the dataset, several endpoint nodes of the same type (e.g. Serotonin syndrome, Rhabdomyolysis, Lactic acidosis) share identical static audited attributes (e.g. `severity_weight`, observability) and comparable topological positions, and were therefore indistinguishable to the model except through such positional cues. To break this symmetry, a learnable embedding table is maintained per node type, indexed by each node's fixed local index, and added to the projected input features before the first convolutional layer, following the same principle underlying position aware node embeddings in .

This connects directly to the model's handling of oversmoothing. Using multiple message-passing layers risks oversmoothing the embedding signal, where repeated aggregation causes node representations to converge toward indistinguishability. To mitigate this, several oversmoothing countermeasures were applied: residual connections, Jumping Knowledge, DropEdge, and PairNorm. These methods were evaluated both individually and in combination with the HCR enrichment described in section 5.2.3.

5.2.3. HCR application

In addition to the purely structural GNN pathway, approch presented in the thesis incorporates a statistical evidence side-channel - HCR described in details in section 3.2. This channel is motivated by the observation that some parent-endpoint dependencies are well captured by simple, well-understood pairwise statistics computed directly from the training cohort, and that combining such statistics with the deep GNN representation captures the strong signal from direct parents and smoothed signal from further layers. There were run couple of experiments with HCR configurations and GNNs. However, in general the approach was about computation of contingency-table statistics for each target endpoint and direct parents or pre-direct parents in the audited DAG (restricted to nodes with inferred binary, count, or continuous value types). The statistics computed was as followed per (parent, endpoint) pair: the φ-coefficient, normalized mutual information, a tanh-scaled log odds ratio, risk difference, and a tanh-scaled log lift, plus prevalence and support terms. There are nine to forty-one channels depending on configuration.

As these coefficients are computed with the patient's own endpoint label as part of the statistic, a naive computation would leak the label into the feature. This was addressed with an exact leave-one-out correction: every training patient's coefficient is computed from all other training patients only, using a closed-form re-weighting of the population statistic.

Continuous and count-valued parents are mapped to comparable pseudo-uniform scores via a train-only empirical CDF transform (with deterministic, patient-ID-seeded jitter for count variables), and Legendre polynomial bases are used to capture non-linear dependence on continuous parents.

Three integration modes were supported:

a) linear approach - a single scalar HCR score added directly to the GNN logit,
b) nonlinear encoder - parent-endpoint evidence vectors pooled and passed through a shared MLP
c) separate encoders per parent node type - every direct parent passed through a separated MLP
d) triple relations - relations not only focused on direct parents, but direct parents and grandparents.

The pooling in HCRPairEncoder masks out padding positions (endpoints with fewer parents than the batch's maximum) and returns a neutral zero vector for endpoints with no eligible parents, avoiding division by zero.

5.2.4. Training setup

Training iterates over patient graphs batched via PyTorch Geometric's DataLoader. The experiment were run with targeted binary cross-entropy and focal loss with class-balance-aware weighting. Weight and focal alpha were computed from the training split's endpoint prevalence, since adverse events are rare and highly imbalanced. Model selection uses ReduceLROnPlateau on validation performance, and evaluation is reported separately per endpoint and in aggregate, using AUROC, AUPRC, and prevalence-normalized lift, alongside the representation-level over-smoothing diagnostics (MAD, cosine similarity, Dirichlet energy) used to relate architectural choices. Due to rare events the main metric for model choice was AUPRC.

6. Experiments results

6.1. Endpoint prediction

The first aim of experiments was to choose the best architecture for the problem. The following architecture was tested RGCN, SAGE, Transformer. All of architecture is designed for homogenous graphs. As pharmaceutical data that we used was structured in heterogenous graph, all of architecture was implemented in heterogenous convolutional layer - heteroconv.

6.1.1. Synthethic dataset

The results for the presented experiments are following.

Tabela 6.1. Transformers with HCR and residual

				Validation				Test			
L	Residual	HCR	JK	AUC	AUPRC	Lift	%AUC	AUC	AUPRC	Lift	%AUC
1	no	—	—	0.621	0.1252	2.30	80	0.618	0.1192	2.19	78
3	no	—	—	0.546	0.0850	1.56	30	0.550	0.0752	1.38	33
3	no	—	cat	0.606	0.1383	2.54	71	0.615	0.1235	2.27	76
3	no	linear 41D	—	0.599	0.1214	2.23	66	0.610	0.1066	1.96	73
3	no	linear 41D	cat	0.608	0.1379	2.53	72	0.616	0.1246	2.29	77
3	yes	—	—	0.624	0.1353	2.48	82	0.618	0.1221	2.24	78
3	yes	—	cat	0.624	0.1357	2.49	82	0.620	0.1236	2.27	80
3	yes	linear 41D	cat	**0.629**	0.1336	2.45	**85**	0.620	0.1243	2.28	80
3	yes	nonlin. 41D	—	0.622	0.1347	2.47	81	0.622	0.1231	2.26	81
3	yes	nonlin. bin.	—	0.624	0.1348	2.47	82	0.622	0.1229	2.26	81
3	yes	typed 41D	—	0.619	0.1342	2.47	79	0.611	0.1211	2.22	74
6	no	—	—	0.526	0.0671	1.23	17	0.524	0.0635	1.17	16
6	no	—	cat	0.617	**0.1415**	**2.60**	78	0.621	**0.1285**	**2.36**	80
6	no	linear 41D	—	0.616	0.1206	2.21	77	0.608	0.1104	2.03	72
6	no	linear 41D	cat	0.615	0.1409	2.59	77	0.623	0.1272	2.34	82
6	yes	—	—	0.624	0.1359	2.50	82	0.621	0.1225	2.25	80
6	yes	—	cat	0.624	0.1352	2.48	82	0.622	0.1239	2.28	81
6	yes	linear 41D	cat	0.626	0.1360	2.50	83	**0.628**	0.1251	2.30	**85**
6	yes	nonlin. 41D	—	0.620	0.1325	2.43	80	0.615	0.1208	2.22	76
6	yes	nonlin. bin.	—	0.604	0.1179	2.16	69	0.604	0.1078	1.98	69
6	yes	typed 41D	—	0.621	0.1338	2.46	80	0.617	0.1230	2.26	78

Tabela 6.2. Transformers with oversmoothing methods applied

			Validation				Test			
L	Residual	Method	AUC	AUPRC	Lift	%AUC	AUC	AUPRC	Lift	%AUC
1	no	—	0.621	0.1252	2.30	80	0.618	0.1192	2.19	78
3	no	—	0.546	0.0850	1.56	30	0.550	0.0752	1.38	33
3	no	JK cat	0.606	0.1383	2.54	71	0.615	0.1235	2.27	76
3	no	JK max	0.622	0.1372	2.52	81	0.619	0.1240	2.28	79
3	no	DropEdge	0.546	0.0800	1.47	30	0.547	0.0726	1.33	31
3	no	JK cat + DropEdge	0.623	0.1386	2.55	81	0.623	0.1249	2.30	82
3	no	JK max + DropEdge	0.626	0.1389	2.55	84	0.622	0.1219	2.24	81
3	yes	—	0.624	0.1353	2.48	82	0.618	0.1221	2.24	78
3	yes	JK cat	0.624	0.1357	2.49	82	0.620	0.1236	2.27	80
3	yes	JK max	0.626	0.1370	2.52	83	0.620	0.1234	2.27	79
3	yes	DropEdge	0.626	0.1341	2.46	84	0.622	0.1237	2.27	81
3	yes	JK cat + DropEdge	**0.629**	0.1362	2.50	**85**	**0.624**	0.1248	2.29	**83**
3	yes	JK max + DropEdge	0.625	0.1359	2.50	83	0.622	0.1238	2.28	81
6	no	—	0.526	0.0671	1.23	17	0.524	0.0635	1.17	16
6	no	JK cat	0.617	**0.1415**	**2.60**	78	0.621	**0.1285**	**2.36**	80
6	yes	—	0.624	0.1359	2.50	82	0.621	0.1225	2.25	80
6	yes	JK cat	0.624	0.1352	2.48	82	0.622	0.1239	2.28	81

Tabela 6.3. Oversmoothing metrics

L	Residual	Metoda	MAD	CosSim	Rank	DirE	Std	E_L/E_0	$\mathrm{MAD}_L/\mathrm{MAD}_0$
1	no	—	0.882	0.571	0.229	0.178	0.168	0.115	1.585
3	no	—	0.799	0.630	0.125	0.041	0.083	0.057	0.898
3	no	JK cat	0.837	0.603	0.070	0.098	0.088	0.018	1.138
3	no	JK max	0.766	0.669	0.208	0.271	0.197	0.019	1.088
3	no	DropEdge	0.750	0.652	0.135	0.053	0.086	0.042	0.871
3	no	JK cat + DropEdge	0.827	0.610	0.073	0.073	0.082	0.020	1.093
3	no	JK max + DropEdge	0.727	0.687	0.190	0.175	0.153	0.024	1.161
3	yes	—	0.774	0.689	0.180	0.189	0.182	0.098	1.112
3	yes	JK cat	0.826	0.630	0.104	0.219	0.174	0.038	1.024
3	yes	JK max	0.630	0.816	0.315	0.337	0.219	0.059	1.039
3	yes	DropEdge	0.784	0.682	0.169	0.110	0.143	0.102	1.102
3	yes	JK cat + DropEdge	0.826	0.633	0.099	0.215	0.177	0.047	0.984
3	yes	JK max + DropEdge	0.678	0.799	0.271	0.279	0.206	0.066	1.094
6	no	—	0.808	0.634	0.125	0.049	0.094	0.080	0.781
6	no	JK cat	0.813	0.611	0.037	0.053	0.056	0.012	1.068
6	yes	—	0.756	0.702	0.133	0.096	0.135	0.056	1.055
6	yes	JK cat	0.823	0.652	0.056	0.163	0.127	0.017	0.939

6.1.2. Proxy dataset

Tabela 6.4. Results under full observability on the proxy dataset. The epoch was selected on the validation set. -- in the HCR column denotes a model without the HCR branch. Lift is AUPRC divided by the prevalence baseline. Jumping knowledge was not used in any configuration.

			Validation			Test		
L	Residual	HCR	AUC	AUPRC	Lift	AUC	AUPRC	Lift
1	no	—	0.757	0.188	7.92	0.664	0.120	6.49
1	no	nonlin. 9D	0.763	0.207	8.75	0.657	0.120	6.47
1	no	typed 41D	0.759	0.194	8.17	0.663	0.116	6.29
1	yes	—	0.756	0.196	8.26	0.669	0.106	5.72
1	yes	nonlin. 9D	0.756	0.201	8.49	0.670	0.135	7.31
1	yes	typed 41D	0.754	0.185	7.79	0.671	0.124	6.71
3	no	—	0.756	0.199	8.38	**0.722**	0.152	8.24
3	no	nonlin. 9D	0.805	0.275	11.62	0.686	0.169	9.14
3	no	typed 41D	0.812	0.243	10.26	0.678	0.154	8.32
3	yes	—	0.799	0.225	9.48	0.685	0.166	8.97
3	yes	nonlin. 9D	0.814	0.235	9.90	0.714	0.183	9.92
3	yes	typed 41D	0.802	0.255	10.76	0.678	0.151	8.16
6	no	—	0.610	0.121	5.11	0.585	0.077	4.14
6	no	nonlin. 9D	0.782	0.210	8.86	0.680	0.174	9.42
6	no	typed 41D	0.794	0.235	9.89	0.680	0.126	6.84
6	yes	—	0.817	0.255	10.77	0.728	0.173	9.36
6	yes	nonlin. 9D	**0.832**	**0.310**	**13.06**	**0.760**	0.179	9.70
6	yes	typed 41D	0.815	0.284	11.99	0.744	**0.239**	**12.91**

Tabela 6.5. Results under restricted (*bedside*) observability, $L = 3$.

		Validation		Test		
Residual	HCR	AUC	Lift	AUC	AUPRC	Lift
no	—	0.568	4.20	0.556	0.062	3.36
no	lin. 41D	0.730	10.09	0.681	0.150	8.10
no	nonlin. 41D	0.751	9.39	0.678	0.154	8.35
no	typed 41D	0.753	9.45	0.679	0.135	7.31
yes	—	0.568	4.39	0.555	0.076	4.14
yes	lin. 41D	0.732	10.78	0.683	0.168	9.10
yes	nonlin. 41D	0.754	10.24	0.679	0.164	8.89
yes	typed 41D	0.754	10.18	0.678	0.173	9.33

Tabela 6.6. Representation smoothing metrics under full observability, measured on the validation set. By the conventional reading, lower MAD and higher cosine similarity indicate stronger smoothing. Validation AUC is repeated for comparison. Note that residual configurations appear more smoothed by every metric while performing better.

L	Residual	HCR	MAD	Cos. sim.	Cos. sim. (ep.)	Dirichlet	AUC
1	no	—	0.7691	0.6216	0.8246	0.2706	0.757
1	no	nonlin. 9D	0.7629	0.6142	0.7844	0.2495	0.763
1	no	typed 41D	0.7684	0.6191	0.8148	0.2706	0.759
1	yes	—	0.7363	0.7042	0.7606	0.5724	0.756
1	yes	nonlin. 9D	0.7448	0.6946	0.7060	0.5048	0.756
1	yes	typed 41D	0.7410	0.7026	0.7899	0.5022	0.754
3	no	—	0.6515	0.6798	0.9582	0.1175	0.756
3	no	nonlin. 9D	0.7067	0.6641	0.8365	0.0904	0.805
3	no	typed 41D	0.7235	0.6314	0.7333	0.0972	0.812
3	yes	—	0.4842	0.8319	0.7772	0.3712	0.799
3	yes	nonlin. 9D	0.5517	0.7782	0.7752	0.3905	0.814
3	yes	typed 41D	0.5113	0.8275	0.7132	0.4044	0.802
6	no	—	0.7060	0.6530	0.8604	0.1024	0.610
6	no	nonlin. 9D	0.7101	0.6495	0.7815	0.0892	0.782
6	no	typed 41D	0.7222	0.6408	0.8657	0.0949	0.794
6	yes	—	0.5634	0.7975	0.7949	0.3143	0.817
6	yes	nonlin. 9D	0.5030	0.8174	0.6658	0.2636	0.832
6	yes	typed 41D	0.5623	0.8095	0.8472	0.3930	0.815

7. Experiments results

8. Link prediction

8.1. Endpoint prediction

9. Podsumowanie

Podsumowanie i krytyczna analiza osiągniętych rezultatów. Ocena, czy osiągnięto założony cel pracy. Dyskusja co można było zrobić lepiej i propozycja dalszych prac w celu usprawieniania opracowanego rozwiązania.

Spis rysunków

Spis tabel